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Azure Automation troubleshooting documentation

Welcome to Azure Automation troubleshooting. These articles explain how to determine, diagnose, and fix various issues that you might encounter when you use Azure Automation. In the navigation pane on the left, browse through the article list or use the search box to find issues and solutions.

Runbook not working as expected

HOW-TO GUIDE

- Runbook jobs get suspended
- Troubleshoot error codes during runbook execution
- Troubleshoot issues with python packages
- Troubleshoot issues with runbook execution start time
- Troubleshoot runbook execution issues
- Troubleshoot runbook execution issues when using PowerShell

Common runbook creation problems on Hybrid Runbook Worker

07/18/2025

This article describes common issues you might experience when you set up runbooks on a hybrid runbook worker.

> **Note**
>
> The agent-based user hybrid runbook worker (Windows and Linux) was retired on August 31, 2024, and is no longer supported. For more information, see migration guidance.

Troubleshoot tools

- Connectivity problems are a common cause of issues with Hybrid Runbook Workers. Use the Test Cloud Connectivity tool to verify that your environment is correctly configured.
- Run the offline version of the agent registration script to troubleshoot hybrid worker prerequisites. Although the script includes some checks specific to update management, most of the requirements also apply to hybrid workers.

Common issues and solutions

Review the following table to resolve other common issues.

Expand table

Issue/ Error	Solution
VM extension-based Hybrid Runbook Worker issues	Troubleshoot VM extension-based Hybrid Runbook Worker issues in Automation
Error: Job action 'Activate' cannot be run.	See the hybrid runbook worker troubleshooting guide under "Runbook execution fails".
Error: No certificate was found in the certificate store.	To resolve this issue, follow the "No certificate was found" section of the hybrid worker troubleshooter.
Error: Machine is already registered.	Follow the troubleshooting guide for "Unable to add a hybrid runbook worker".

Reference

- Automation hybrid runbook workers
- Deploy an extension-based Windows or Linux User Hybrid Runbook Worker in Azure Automation
- Migrate the existing agent-based hybrid workers to extension-based hybrid workers

Hybrid Runbook Worker job isn't working as expected

07/16/2025

This article provides guidance for troubleshooting and resolving issues that affect Hybrid Runbook Worker in Azure Automation.

> **Note**
>
> Azure Automation enables the recovery of runbooks that are deleted in the past 29 days. For more information, see Restore deleted runbook.

Troubleshoot connectivity issues

Connectivity problems are a common cause of issues that affect runbooks. Use the Test Cloud Connectivity tool to verify that your environment is correctly configured.

General troubleshooting

Expand table

Issue	Resolution
Runbooks behave differently on a hybrid worker than in Azure Automation.	See Runbook permissions for information about authentication differences.
Error: No certificate was found.	Follow the "No Certificate Found" section in the troubleshooting guide.
You need to troubleshoot a custom runbook.	See Troubleshoot runbook issues.
You need to check job status.	Review job details and statuses.
Hybrid worker doesn't run jobs or is unresponsive.	Troubleshoot by using Hybrid Runbook Worker diagnostics.
Runbooks suddenly stop working.	Make sure that you migrated to managed identity and that webhooks aren't expired.
Need help with passing	See Start a runbook from a webhook.

Issue	Resolution
parameters into a webhook.	
You want to use both `Az` and `AzureRM` modules in runbook.	This scenario isn't supported. Use only Az modules in runbooks.
Can't start or schedule a runbook.	Verify that the runbook is in a published state.
Runbook is suspended or failed unexpectedly.	Review job statuses. Add logging to the runbook by using output streams. If the job fails three times, check the automation limits and consider using a Hybrid Worker.

Windows Hybrid Runbook Worker issues

Expand table

Issue	Resolution
Event 4502 appears in the Operations Manager log.	See Event 4502.
Script using `Connect-MsolService` can't connect to Microsoft 365.	See Sandbox can't connect to Microsoft 365.
Machine is running but not sending heartbeat data.	See Hybrid worker not reporting.

Linux Hybrid Runbook Worker issues

Expand table

Issue	Resolution
Unexpected password prompt appears when using `sudo`.	See Linux runbook worker prompts for password.
Log file shows "The specified class does not exist."	See Class does not exist error.
Linux job is stuck in **Running** state	1. Switch to `sudo` permissions: `sudo su` 2. Make sure that the `hwd` service is running: `systemctl status hwd.service` 3. Open the following file in Hybrid Worker: `vi /lib/systemd/system/hwd.service` 4. Update the setting from `CPUQuota=25%` to `CPUQuota=` to make the usage unrestricted, as shown in the following example: `[Unit]`

Issue	Resolution

	`Description=HW Service`
	`After=network.target`
	`[Service]`
	`Type=simple`
	`ExecStart=/usr/bin/python3 .../automationWorkerStarterScript.py`
	`TimeoutStartSec=5`
	`Restart=always`
	`RestartSec=10s`
	`TimeoutStopSec=600`
	`CPUQuota=`
	`KillMode=process`
	`[Install]`
	`WantedBy=multi-user.target`
	
	5. Restart the `hwd` service:
	`systemctl daemon-reload`
	`systemctl restart hwd.service`

Other error messages

Expand table

Error	Resolution
"The subscription cannot be found"	This error usually means that the runbook isn't using a managed identity. Follow the steps in Unable to find subscription.
"Strong authentication enrollment is required."	See Authentication to Azure failed due to MFA.
"No permission" or similar error	Make sure that the managed identity has appropriate permissions.

Reference

- Automation Hybrid Runbook Worker overview
- Deploy a Windows Hybrid Runbook Worker
- Deploy a Linux Hybrid Runbook Worker
- Run runbooks on a Hybrid Runbook Worker

Runbook jobs are suspended in Azure Automation

06/24/2025

This article provides solutions to an issue where runbook jobs are suspended in Azure Automation.

> **Note**
>
> Azure Automation enables recovery of runbooks deleted in the last 29 days. You can restore the deleted runbook by running a PowerShell script as a job in your Automation account. For more information, see Restore a deleted runbook.

Symptoms

Runbook jobs might be suspended after three failed start attempts.

Cause 1: Exceeding memory or network socket limits in an Azure sandbox

- Memory limits

  A job might fail if it uses more than 400 MB of memory.

- Network socket limits

  An Azure sandbox is limited to 1,000 concurrent network sockets.

For more information, see Azure Automation limits.

Here are some suggestions for working within memory limits:

- Split the workload among multiple runbooks.
- Reduce the amount of data processed in memory.
- Avoid writing unnecessary output from your runbooks.
- Consider the number of checkpoints written in your PowerShell workflow runbooks.

Here are some actions that can reduce the memory footprint of your runbook during runtime:

- Use the `clear` method, such as `$myVar.clear`, to clear variables.

- Use the `[GC]::Collect` command to run the garbage collection immediately.

Cause 2: Module incompatibility

Module dependencies might be incorrect. In this case, your runbook typically returns a "Command not found" or "Cannot bind parameter" error message.

To resolve this issue, update Azure PowerShell modules in Azure Automation.

Cause 3: Not authenticated with Microsoft Entra ID in an Azure sandbox

Azure Automation runbooks that attempt to call executables or subprocesses within an Azure sandbox environment can't use Microsoft Authentication Library (MSAL) to authenticate with Microsoft Entra ID.

To resolve this issue, use a managed identity for the Automation account. When you authenticate this runbook with Microsoft Entra ID by using a managed identity, ensure the following things:

- The Microsoft Graph PowerShell module is available in your Automation account.
- The managed identity is granted the permissions required to execute the runbook automation tasks.

If your runbook can't call an executable or subprocess in an Azure sandbox, run the runbook on a hybrid worker. Hybrid workers aren't constrained by the memory and network limits of Azure sandboxes.

References

- Understand how Azure Resource Manager throttles requests
- Troubleshooting API throttling errors

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Troubleshoot error codes during runbook execution in Azure Automation

06/27/2025

When using runbooks in Azure Automation, you might encounter issues due to runtime or PowerShell exceptions or specific error messages. This article lists some issues and provides solutions to them.

> **Note**
>
> Azure Automation enables recovery of runbooks deleted in the last 29 days. You can restore the deleted runbook by running a PowerShell script as a job in your Automation account. For more information, see Restore a deleted runbook.

Troubleshoot error messages

Review the following table to resolve runbook execution-related error messages:

Expand table

Error	Description	Solution
Unable to create a new Automation job in the West Europe region.	This issue occurs due to the scalability limits of the Automation service in the West Europe region.	To resolve this issue, follow the steps in Unable to create new Automation job in West Europe region.
`The subscription cannot be found.`	This issue can occur when the runbook doesn't use a managed identity to access Azure resources.	To resolve this issue, follow the steps in Unable to find the Azure subscription.
`Strong authentication enrollment is required.`	If you have multifactor authentication on your Azure account, you can't use a Microsoft Entra user to authenticate to Azure.	To resolve this issue, see Authentication to Azure fails because multifactor authentication is enabled.
Runbook fails with "No permission" or some variation.	The managed identity might not have the same permissions to Azure resources as your current account.	Make sure that your managed identity has permission to access any resources used in your script.

Error	Description	Solution
Error: "429: The request rate is currently too large. Please try again."	This error can occur when retrieving job output from a runbook that has many verbose streams.	To resolve this error, see The request rate is too large.
`400 Bad Request.`	A "400 Bad Request" error can occur for several reasons. First, verify if the runbook works outside of Azure Automation. Typically, the runbook code doesn't work with this error due to: - Missing values. - Wrong values being passed. - Older modules being used. - A referenced resource being disabled; for example, the runbook calls a disabled or expired webhook.	To resolve this error, see 400 Bad Request status when calling a webhook.
`Resource not found.`	Check the values provided for the resource.	The three values to check are: - Resource name - Resource group name - Subscription Confirm that the runbook has permissions in the subscription that contains the resource. To change the subscription, you can use the `Set-AzContext` or `az account set` command together with the `-DefaultProfile` parameter. Many commands provide a subscription parameter that allows a different subscription to be used than the current one. Sign in to the Microsoft Azure portal. Find the resource being used and examine the resource name, resource group, and subscription.

If none of these solutions addresses your issue, see Troubleshoot runbook issues.

> **Note**
>
> Before opening a case, follow the steps in Data to collect when opening a case for Azure Automation. This process will help us resolve your case as quickly as possible.

References

- Start a runbook in Azure Automation
- Run Automation runbooks on a Hybrid Runbook Worker

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Troubleshoot issues with python packages in Azure Automation

06/27/2025

The article shows how to import, manage, and use Python packages in Azure Automation running on the Azure sandbox environment and Hybrid Runbook Workers. Python packages should be downloaded on Hybrid Runbook workers for successful job execution. To help simplify runbooks, you can use Python packages to import the modules you need.

> **Note**
>
> Azure Automation enables the recovery of runbooks deleted in the last 29 days. You can restore the deleted runbook by running a PowerShell script as a job in your Automation account. For more information, see Restore deleted runbook.

Import Python 2 packages

1. In your Automation account, select **Python packages** under **Shared Resources**. Select **+ Add a Python package**.
2. On the **Add Python Package** page, select a local package to upload. The package can be a **.whl** or **.tar.gz** file.
3. Enter the name and select the **Runtime version** as "2.x.x".
4. Select **Import**.

After a package is imported, it's listed on the **Python packages** page in your Automation account. To remove a package, select the package and select **Delete**.

Import packages with dependencies

Azure Automation doesn't resolve dependencies for Python packages during the import process. Use one of the following methods to import a package with all its dependencies.

Method 1: Manually download

On a Windows 64-bit machine with Python 2.7 and the package installer for Python (pip), download a package and all its dependencies by running the following command:

```
Console

C:\Python27\Scripts\pip2.7.exe download -d <output-directory> <package-name>
```

Once the packages and all its dependencies are downloaded, you can import them into your Automation account.

Method 2: Use a runbook

To get a runbook, import Python 2 packages from pypi into Azure Automation account.

When you start the runbook, ensure the following things:

- The **Run on** option under **Run Settings** is set to **Azure**.
- The runbook is started with the following parameters and each parameter is defined with a `switch`.
  - -s <subscriptionId>
  - -g <resourceGroup>
  - -a <automationAccount>
  - -m <modulePackage>

  The runbook allows you to specify which package to download. For example, set the `-m` parameter to `Azure` to download all Azure modules and all dependencies (approximately 105 packages).
- The runbook requires a managed identity for the Automation account to work.

After the runbook execution is complete, you can check the **Python packages** under **Shared Resources** in your Automation account to verify that the package is imported correctly.

Default Python 3 packages

To support Python 3.8 runbooks in the Automation service, some Python packages are installed by default. For more information, see Default Python packages. The default version can be overridden by importing Python packages into your Automation account. The imported version is preferred in your Automation account. To import a single package, see Import a Python 3 package. To import a package with multiple packages, see Import a Python 3 package with dependencies.

> **Note**

Python 3.10 (preview) doesn't have default packages installed.

Import a Python 3 package

1. In your Automation account, select **Python packages** under **Shared Resources**. Then, select + **Add a Python package**.
2. On the **Add Python Package** page, select a local package to upload. The package can be a **.whl** or **.tar.gz** file for Python 3.8 and a **.whl** file for Python 3.10 (preview).
3. Type a name and select the **Runtime Version** as Python 3.8 or Python 3.10 (preview).
4. Select **Import**.

After a package is imported, it's listed on the Python packages page in your Automation account. To remove a package, select the package and select **Delete**.

Import a Python 3 package with dependencies

You can import a Python 3.8 package and its dependencies by importing the Python script import_py3package_from_pypi.py into a Python 3.8 runbook. Ensure that a managed identity is enabled for your Automation account and has Automation Contributor access to import the package successfully.

Import the script into a runbook

For more information about importing the runbook, see Import a runbook from the Azure portal. Copy the file from GitHub to storage that the Azure portal can access before you run the import.

The **Import a runbook** page defaults the runbook name to match the name of the script. If you have access to the field, you can change the name. **Runbook type** might default to **Python 2.7**. If it does, make sure to change it to **Python 3.8**.

Execute the runbook to import the package and dependencies

After creating and publishing the runbook, run it to import the package. For more information about executing a runbook, see Start a runbook in Azure Automation.

The script (import_py3package_from_pypi.py) requires the following parameters.

Expand table

Parameter	Description
`subscription_id`	Subscription ID of the Automation account
`resource_group`	Name of the resource group where the Automation account is defined
`automation_account`	Automation account name
`module_name`	Name of the module to import from *pypi.org*
`module_version`	Version of the module

Parameter value should be provided as a single string in the following format:

`-s <subscription_id> -g <resource_group> -a<automation_account> -m <module_name> -v <module_version>`

Scenario: Runbook fails with "Parameter length exceeded" error

Your runbook that uses those parameters fails with the following error:

> Total Length of Runbook Parameter names and values exceeds the limit of 30,000 characters. To avoid this issue, use Automation Variables to pass values to runbook.

Cause

Python 2.7, Python 3.8, and PowerShell 7.1 runbooks have a limit on the total length of characters for all parameters provided. The total length of all parameter names and values can't exceed 30,000 characters.

Resolution

To resolve this issue, use Azure Automation Variables to pass values to the runbook or shorten parameter names and values so they don't exceed 30,000 characters in total.

References

- Manage modules in Azure Automation
- Runbook execution in Azure Automation

- Manage Python 2 packages in Azure Automation
- Manage Python 3 packages in Azure Automation

Troubleshoot issues with runbook execution start time in Azure Automation

07/23/2025

Process automation in Azure Automation allows you to create and manage PowerShell, a PowerShell workflow, and graphical runbooks. Automation executes your runbooks based on the logic defined within them.

Service-level agreement (SLA)

The SLA for runbook automation is 30 minutes. It's expected that 99.9 percent of runbooks start within 30 minutes of the planned start time. For more information, see SLAs for Online Services.

The *SLA for the Automation Service - Process Automation* defines the following terms:

- "Delayed Jobs" is the total number of Jobs, for a given Microsoft Azure subscription, that fail to start within 30 minutes of their Planned Start Times.
- "Job" means the execution of a Runbook.
- "Planned Start Time" is a time at which a Job is scheduled to begin executing.
- "Runbook" means a set of actions specified by you to run within Microsoft Azure.

> **Note**
>
> - Azure Automation enables the recovery of runbooks deleted in the last 29 days. You can restore the deleted runbook by running a PowerShell script as a job in your Automation account. For more information, see Restore deleted runbook.
> - Before opening a case, follow the steps in Data to collect when opening a case for Azure Automation. This process helps resolve your case as quickly as possible.

References

- Start a runbook in Azure Automation
- Azure Automation runbook types

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Troubleshoot runbook execution issues in Azure Automation

07/01/2025

Process automation in Azure Automation allows you to create and manage PowerShell, a PowerShell workflow, and graphical runbooks. Azure Automation executes your runbooks based on the logic defined within them. This article provides troubleshooting guidance on runbook execution issues in Azure Automation.

> **Note**
>
> - If you use the Newtonsoft.Json v10, be sure to import this module explicitly so that your PowerShell 5.1 and PowerShell 7.1 runbooks that have a dependency on this version of the module function correctly.
> - PowerShell 7.1 is no longer supported by the parent product PowerShell. We recommend creating PowerShell 7.2 runbooks for long-term support. Learn more about PowerShell 7.2 runbooks.
> - If none of the following solutions resolves your issue, see Data to collect when opening a case for Microsoft Azure Automation before opening a support case.
> - Azure Automation enables the recovery of runbooks deleted in the last 29 days. You can restore the deleted runbook by running a PowerShell script as a job in your Automation account. For more information, see Restore deleted runbook.

Troubleshoot error code 400, 403, and 429

Expand table

Error	Solution
`400 Bad Request: This webhook is expired or disabled`	To resolve this error, see 400 Bad Request status when calling a webhook.
`The remote server returned an error: (403) Forbidden`	To resolve this error, see Access blocked to Azure Storage, or Azure Key Vault, or Azure SQL.
`this.Client.SubscriptionId cannot be null`	To resolve this error, see Runbook fails with "this.Client.SubscriptionId cannot be null." error message.

Error	Solution

`ErrorCode: AuthorizationFailed`
 	
`StatusCode: 403`
	To resolve this error, see Runbooks fail when dealing with multiple subscriptions.

`429: The request rate is currently too large. Please try again.`	To resolve this error, see 429: The request rate is currently too large.

Troubleshoot errors related to cmdlet not recognized, sign-in required, or subscription missing

Expand table

Error	Solution
Runbooks fail with the error `The subscription named <subscription name> cannot be found.`	This error can occur when the runbook doesn't use a managed identity to access Azure resources. To resolve this error, see Unable to find the Azure subscription.
`Your Azure credentials haven't been set up or have expired, please run connect-azureRmAccount to set up your azure credentials.`	This error can occur when you don't use a managed identity. To resolve this error, use a managed identity.
`Command not recognized.`	This error often occurs when modules aren't imported or are out of date. Make sure that dependent modules in your script are imported into Azure Automation and are the correct version. If the module is in your Automation account, there might be an issue loading it into the sandbox. Try adding an explicit `import-module` statement at the beginning of your runbook.
`Forbidden with client authentication scheme 'anonymous'`.	This error occurs when using credentials in an Azure Automation sandbox. To resolve this error, use a managed identity.
`Unable to require token for tenant <tenant-id>`.	This error occurs when using credentials in an Azure Automation sandbox. To resolve this error, use a managed identity.
`Server failed to authenticate the request.`	This error occurs when using credentials in an Azure Automation sandbox. To resolve this error, use a managed identity.

Error	Solution
Errors when using the `Connect-AzAccount` cmdlet.	To resolve this issue, see Sign-in to Azure account failed.
`The term 'Connect-AzAccount' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if the path was included verify that the path is correct and try again.`	To resolve this error, see Term not recognized as the name of a cmdlet, function, or script.
Cmdlet fails in PnP PowerShell runbook on Azure Automation.	To resolve this issue, see Cmdlet fails in PnP PowerShell runbook on Azure Automation.
`<cmdlet name>: The term <cmdlet name> is not recognized as the name of a cmdlet, function, script file, or operable program.`	To resolve this error, see Cmdlet not recognized when executing a runbook.

Troubleshoot issues with suspended runbooks, job failures, stopped runbooks, hybrid workers, and subscriptions

Expand table

Issue	Solution
Runbook is suspended or unexpectedly fails.	Review your Job Statuses for runbook statuses and possible causes. Add more output to identify what happens before the runbook is suspended. Handle any exceptions thrown by your job. Retry your job when certain exceptions occur, such as WebSocket exceptions, to prevent transient network failures from causing runbook failures. If you have issues with a cmdlet, you might find more information with the service you try to use through the cmdlet. For example, the `New-AzAnalysisServicesServer` cmdlet related issues might end up with the Analysis Services team.
Job was tried three times, but it failed.	Check the Automation Limits. If the limitation applies to Azure sandboxes only, consider moving to a hybrid worker.
Runbooks were working but stopped.	Make sure to use a managed identity. If using webhooks to start runbooks, make sure the webhook hasn't expired.

Issue	Solution
Issues when using a hybrid worker.	To resolve these issues, see Hybrid Runbook Worker troubleshooting guide.
Runbook is stuck.	If you can't stop a runbook job in the Azure portal, try stopping it by using the PowerShell cmdlet `Stop-AzureRmAutomationJob` or `Stop-AzAutomationJob`.
Can't start or schedule a runbook.	To resolve this issue, make sure your runbook is published.
Issues when using cmdlets that depend on binaries.	Some cmdlets rely on binaries, such as Microsoft Data Access Components (MDAC) or the Azure Fabric SDK. These cmdlets can't be run in the Azure Automation sandbox and must be executed through a hybrid worker.
Issues when there are multiple subscriptions in a runbook.	To manage Azure resources across several subscriptions with Azure Automation, see Dealing with multiple subscriptions to prevent errors.

Troubleshoot common issues

The following table helps you troubleshoot common issues with runbooks:

Expand table

Issue	Solution
Unable to create a new Automation job in the West Europe region.	This issue occurs because of scalability limits with the Automation service in the West Europe region. To resolve it, follow the steps in Unable to create new Automation job in West Europe region.
Runbook bugs or Azure Automation issues.	To learn how to troubleshoot common scenarios, see Troubleshoot runbook issues.
Runbook output and message issues.	To resolve such issues, see Retrieve runbook output and messages.
PowerShell module issues in Azure Automation.	To resolve such issues, see Update Azure PowerShell modules in Automation.

References

- How to start a runbook in Azure Automation
- PowerShell runbooks

- Set-AzSqlDatabase that sets properties for a database or move an existing database into an elastic pool

Troubleshoot runbook execution issues when using PowerShell in Azure Automation

06/13/2025

This article provides guidance on diagnosing and resolving runbook execution issues that occur when using PowerShell scripts or cmdlets in Azure Automation.

> **Note**
>
> Azure Automation enables recovery of runbooks deleted in last 29 days. You can restore a deleted runbook by running a PowerShell script as a job in your Automation account. For more information, see Restore deleted runbook.

Limitations and known issues with PowerShell runbooks

PowerShell runbooks are built on Windows PowerShell. You can edit their code directly by using the text editor in the Azure portal. You can also use an offline text editor and then import the runbooks into Azure Automation. The PowerShell version is determined by the **Runtime version** specified.

Limitations:

- For the PowerShell 7 runtime version, the module activities aren't extracted for the imported modules.
- The `PSCredential` runbook parameter type isn't supported in PowerShell 7 runtime version.
- PowerShell 7.x doesn't support workflows runbooks. For more information, see PowerShell workflow.
- PowerShell 7.x currently doesn't support signed runbooks.
- Source control integration doesn't support PowerShell 7.2. PowerShell 7.2 runbooks in the source control are created in an Automation account as runtime 5.1.
- Currently, only cloud jobs are supported for the PowerShell 7.2 runtime version.
- Using Az modules and AzureRM modules in the same Automation account isn't supported. For more information, see Update Azure PowerShell modules in Automation.

Known issues:

- Runbook properties defining logging preference isn't supported in PowerShell 7 runtime.

  To work around this issue, explicitly set the preference at the start of the runbook as follows:

  Azure PowerShell

  ```
  $VerbosePreference = "Continue"
  $ProgressPreference = "Continue"
  ```

- When you use the ExchangeOnlineManagement module version 3.0.0 or later, you might experience errors.

  To resolve this issue, make sure that you explicitly upload the `PowerShellGet` and `PackageManagement` modules.

Before troubleshooting

Follow these steps to identify and resolve common errors before further troubleshooting:

1. Verify that your PowerShell script works outside of Azure Automation.
2. Verify that the required modules are imported into your Azure Automation account.

Issue 1: Runbook execution fails due to deserialized object

Symptoms:

When you run a runbook, the following error message might be displayed:

> Cannot bind parameter <ParameterName>.
>
> Cannot convert the <ParameterType> value of type Deserialized <ParameterType> to type <ParameterType>.

Cause:

This error occurs with PowerShell Workflow runbooks because PowerShell Workflow stores complex objects in a deserialized format to persist your runbook state when the workflow is suspended.

Resolution:

To resolve this issue, use one of the following methods:

- If you pipe complex objects from one PowerShell cmdlet to another one, wrap these cmdlets in an `InlineScript` activity.
- Pass the name or value that you need from the complex object instead of passing the entire object.
- Use a PowerShell runbook instead of a PowerShell Workflow runbook.

Issue 2: PowerShell jobs fail with the "Cannot invoke method" error

Symptoms:

When you start a PowerShell job in a runbook that runs in an Azure sandbox, you receive the following error message:

> Exception was thrown - Cannot invoke method. Method invocation is supported only on core types in this language mode.

Cause:

This error might occur because the runbook can't run in the Full Language mode.

Resolution:

To resolve this error, use one of the following methods:

- Use Start-AzAutomationRunbook instead of Start-Job to start the runbook.
- Try running the runbook on a Hybrid Runbook Worker.

Reference

- Troubleshoot runbook issues
- Start a runbook in Azure Automation
- Run runbooks on a Hybrid Runbook Worker
- Update Azure PowerShell modules in Automation

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07/01/2025

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> **Note**
>
> - If you use the Newtonsoft.Json v10, be sure to import this module explicitly so that your PowerShell 5.1 and PowerShell 7.1 runbooks that have a dependency on this version of the module function correctly.
> - PowerShell 7.1 is no longer supported by the parent product PowerShell. We recommend creating PowerShell 7.2 runbooks for long-term support. Learn more about PowerShell 7.2 runbooks.
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Troubleshoot error code 400, 403, and 429

Expand table

Error	Solution
`400 Bad Request: This webhook is expired or disabled`	To resolve this error, see 400 Bad Request status when calling a webhook.
`The remote server returned an error: (403) Forbidden`	To resolve this error, see Access blocked to Azure Storage, or Azure Key Vault, or Azure SQL.
`this.Client.SubscriptionId cannot be null`	To resolve this error, see Runbook fails with "this.Client.SubscriptionId cannot be null." error message.

Error	Solution

`ErrorCode: AuthorizationFailed`
 	
`StatusCode: 403`
	To resolve this error, see Runbooks fail when dealing with multiple subscriptions.

`429: The request rate is currently too large. Please try again.`	To resolve this error, see 429: The request rate is currently too large.

Troubleshoot errors related to cmdlet not recognized, sign-in required, or subscription missing

Expand table

Error	Solution
Runbooks fail with the error `The subscription named <subscription name> cannot be found.`	This error can occur when the runbook doesn't use a managed identity to access Azure resources. To resolve this error, see Unable to find the Azure subscription.
`Your Azure credentials haven't been set up or have expired, please run connect-azureRmAccount to set up your azure credentials.`	This error can occur when you don't use a managed identity. To resolve this error, use a managed identity.
`Command not recognized.`	This error often occurs when modules aren't imported or are out of date. Make sure that dependent modules in your script are imported into Azure Automation and are the correct version. If the module is in your Automation account, there might be an issue loading it into the sandbox. Try adding an explicit `import-module` statement at the beginning of your runbook.
`Forbidden with client authentication scheme 'anonymous'`.	This error occurs when using credentials in an Azure Automation sandbox. To resolve this error, use a managed identity.
`Unable to require token for tenant <tenant-id>`.	This error occurs when using credentials in an Azure Automation sandbox. To resolve this error, use a managed identity.
`Server failed to authenticate the request.`	This error occurs when using credentials in an Azure Automation sandbox. To resolve this error, use a managed identity.

Error	Solution
Errors when using the `Connect-AzAccount` cmdlet.	To resolve this issue, see Sign-in to Azure account failed.
`The term 'Connect-AzAccount' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if the path was included verify that the path is correct and try again.`	To resolve this error, see Term not recognized as the name of a cmdlet, function, or script.
Cmdlet fails in PnP PowerShell runbook on Azure Automation.	To resolve this issue, see Cmdlet fails in PnP PowerShell runbook on Azure Automation.
`<cmdlet name>: The term <cmdlet name> is not recognized as the name of a cmdlet, function, script file, or operable program.`	To resolve this error, see Cmdlet not recognized when executing a runbook.

Troubleshoot issues with suspended runbooks, job failures, stopped runbooks, hybrid workers, and subscriptions

Expand table

Issue	Solution
Runbook is suspended or unexpectedly fails.	Review your Job Statuses for runbook statuses and possible causes. Add more output to identify what happens before the runbook is suspended. Handle any exceptions thrown by your job. Retry your job when certain exceptions occur, such as WebSocket exceptions, to prevent transient network failures from causing runbook failures. If you have issues with a cmdlet, you might find more information with the service you try to use through the cmdlet. For example, the `New-AzAnalysisServicesServer` cmdlet related issues might end up with the Analysis Services team.
Job was tried three times, but it failed.	Check the Automation Limits. If the limitation applies to Azure sandboxes only, consider moving to a hybrid worker.
Runbooks were working but stopped.	Make sure to use a managed identity. If using webhooks to start runbooks, make sure the webhook hasn't expired.

Issue	Solution
Issues when using a hybrid worker.	To resolve these issues, see Hybrid Runbook Worker troubleshooting guide.
Runbook is stuck.	If you can't stop a runbook job in the Azure portal, try stopping it by using the PowerShell cmdlet `Stop-AzureRmAutomationJob` or `Stop-AzAutomationJob`.
Can't start or schedule a runbook.	To resolve this issue, make sure your runbook is published.
Issues when using cmdlets that depend on binaries.	Some cmdlets rely on binaries, such as Microsoft Data Access Components (MDAC) or the Azure Fabric SDK. These cmdlets can't be run in the Azure Automation sandbox and must be executed through a hybrid worker.
Issues when there are multiple subscriptions in a runbook.	To manage Azure resources across several subscriptions with Azure Automation, see Dealing with multiple subscriptions to prevent errors.

Troubleshoot common issues

The following table helps you troubleshoot common issues with runbooks:

Expand table

Issue	Solution
Unable to create a new Automation job in the West Europe region.	This issue occurs because of scalability limits with the Automation service in the West Europe region. To resolve it, follow the steps in Unable to create new Automation job in West Europe region.
Runbook bugs or Azure Automation issues.	To learn how to troubleshoot common scenarios, see Troubleshoot runbook issues.
Runbook output and message issues.	To resolve such issues, see Retrieve runbook output and messages.
PowerShell module issues in Azure Automation.	To resolve such issues, see Update Azure PowerShell modules in Automation.

References

- How to start a runbook in Azure Automation
- PowerShell runbooks

- Set-AzSqlDatabase that sets properties for a database or move an existing database into an elastic pool